AI for Systems

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DATA SCIENCE CONSULTANT

@matthewrenze matthewrenze.com

Overview

AI Systems

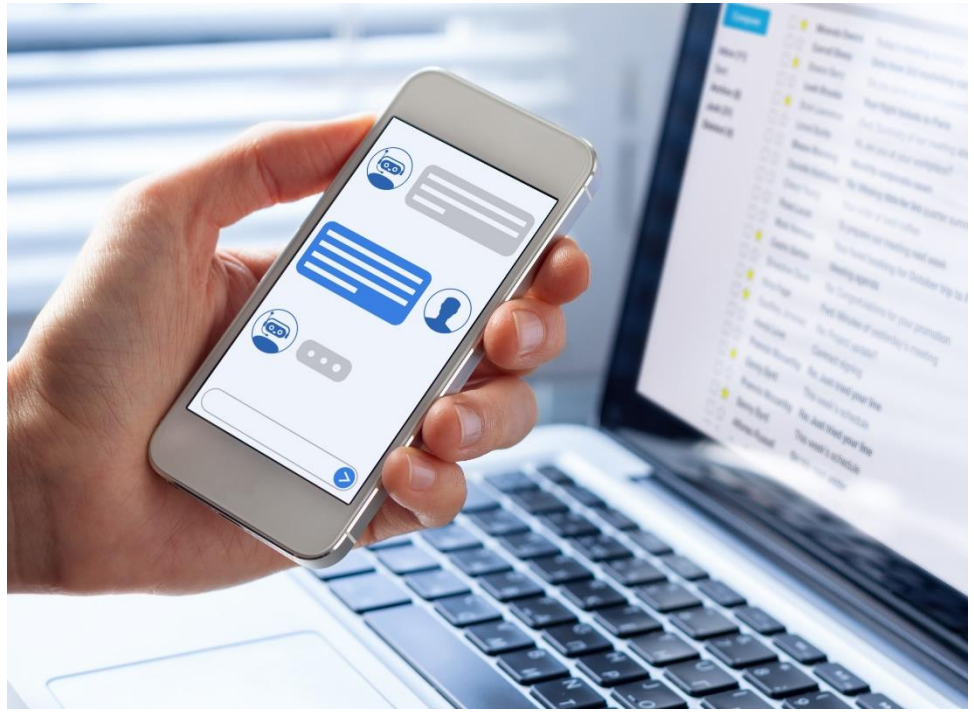
The Process

Demo

Systems

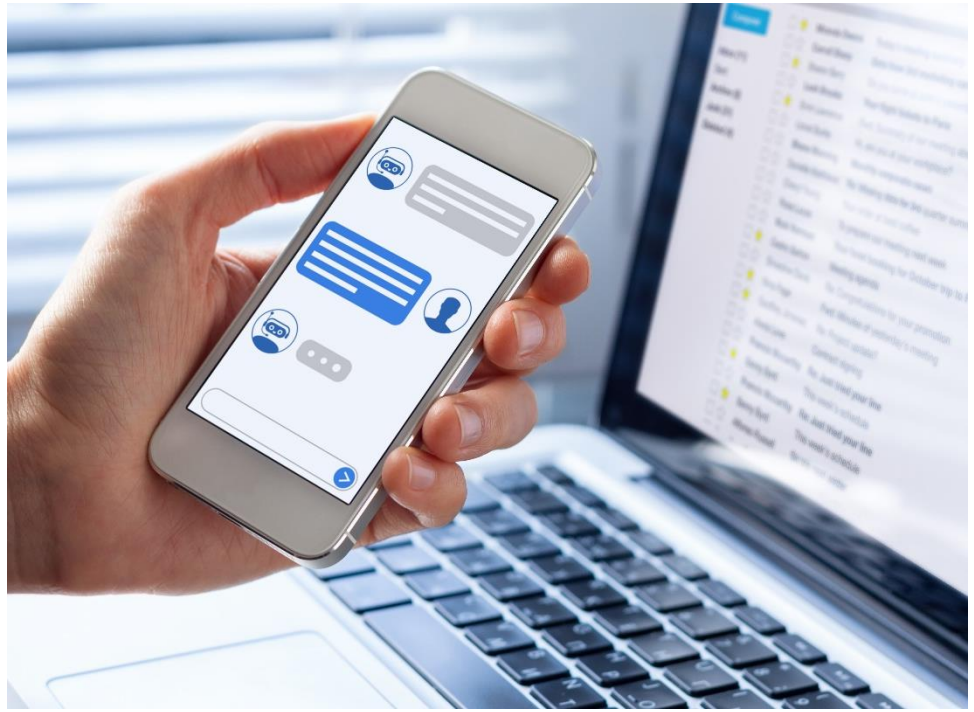
Systems

AI Application



Systems

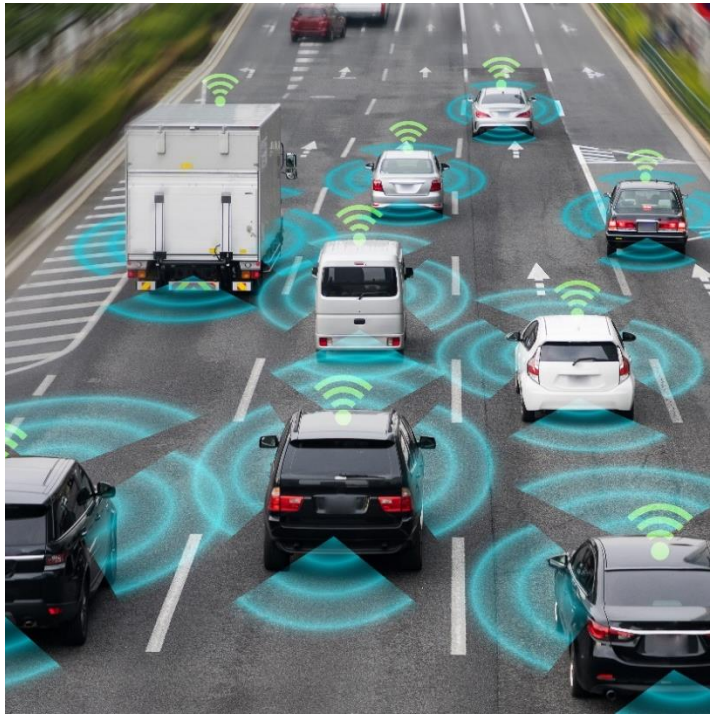
AI Application



vs.

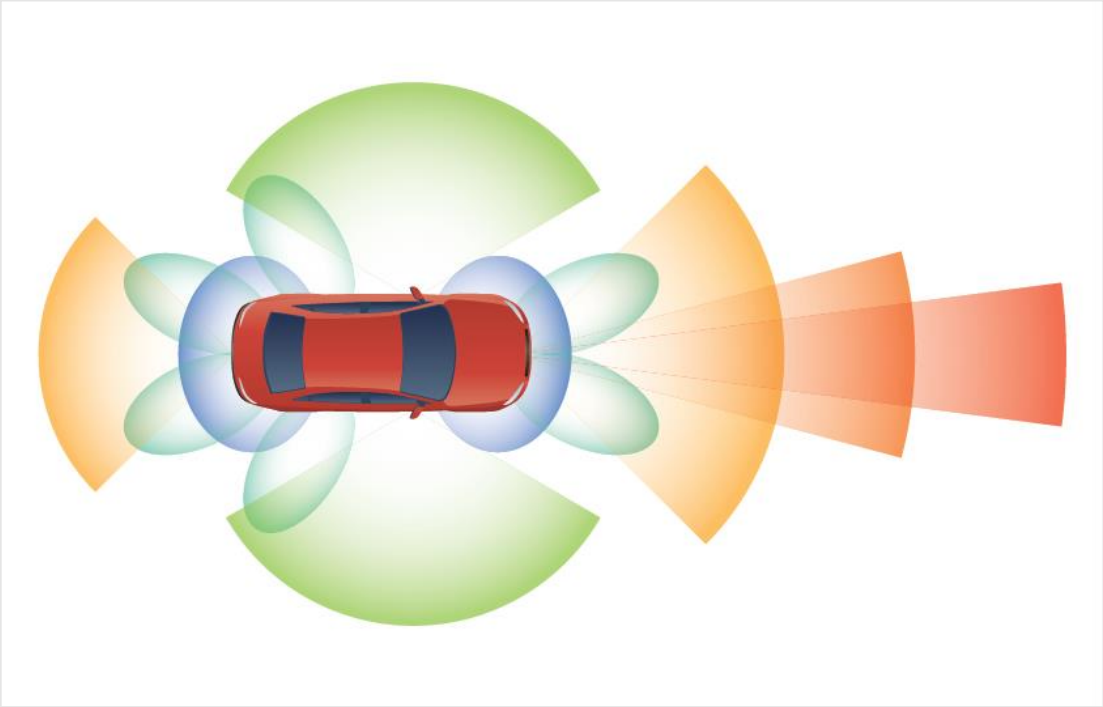
Cyber-physical System

Systems



Systems

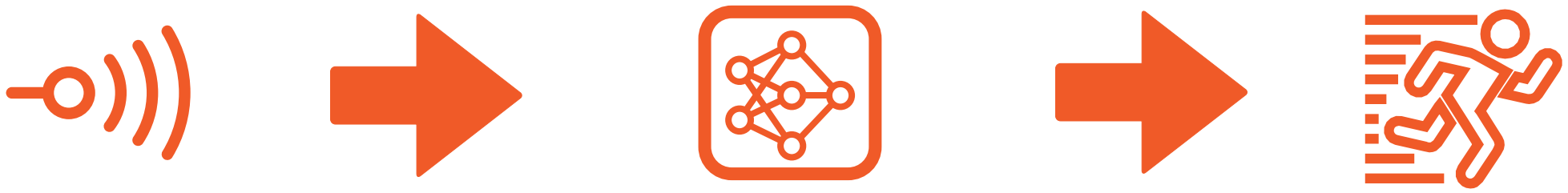
Sensors



Actuators

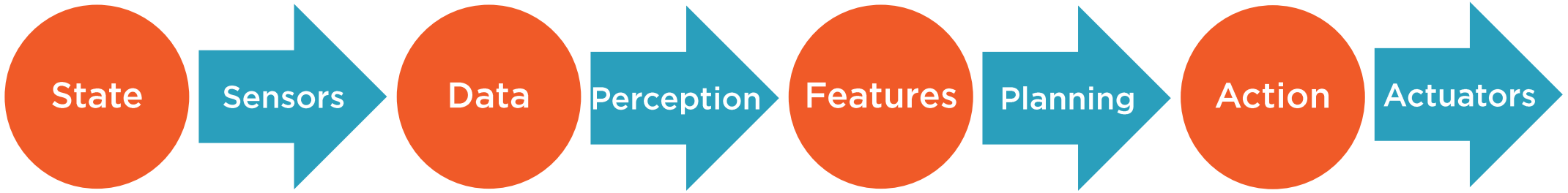
The Process

It's All Just a Function

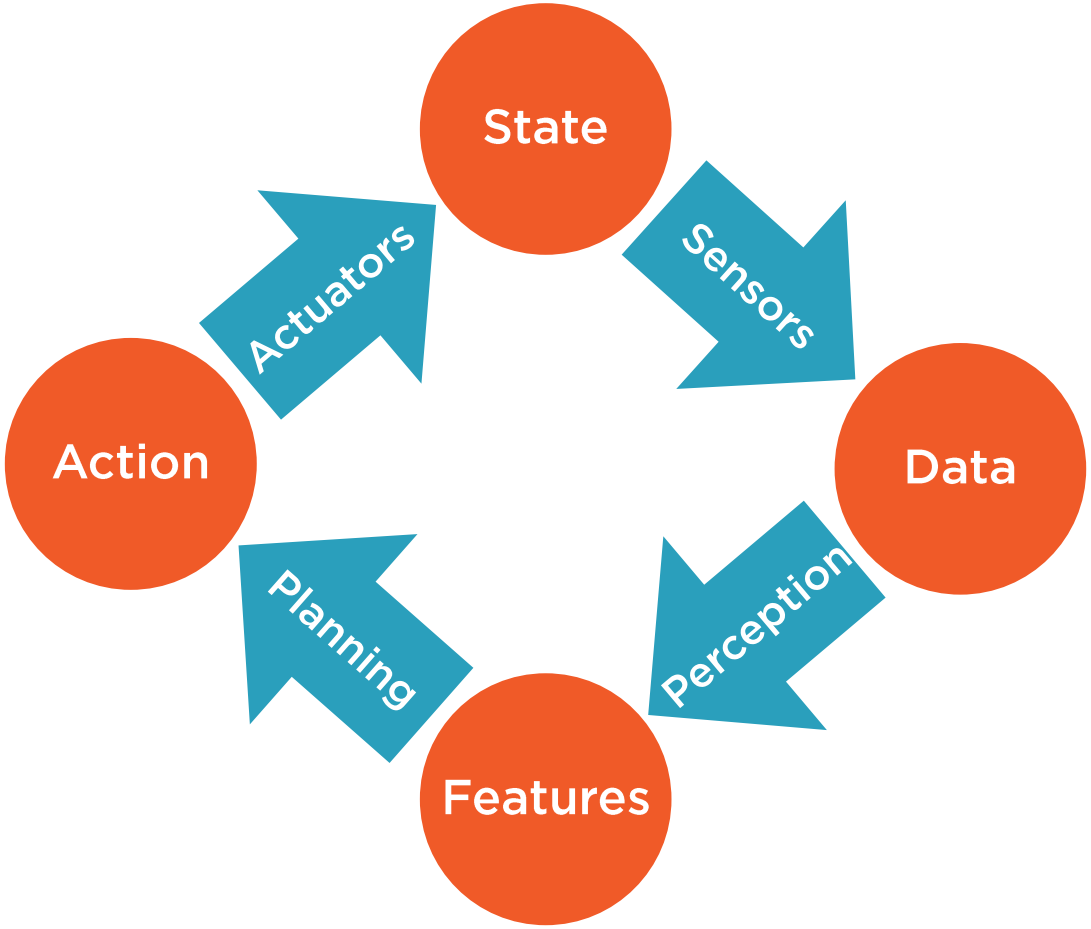


input $\longrightarrow$ $f(x)$ $\longrightarrow$ output

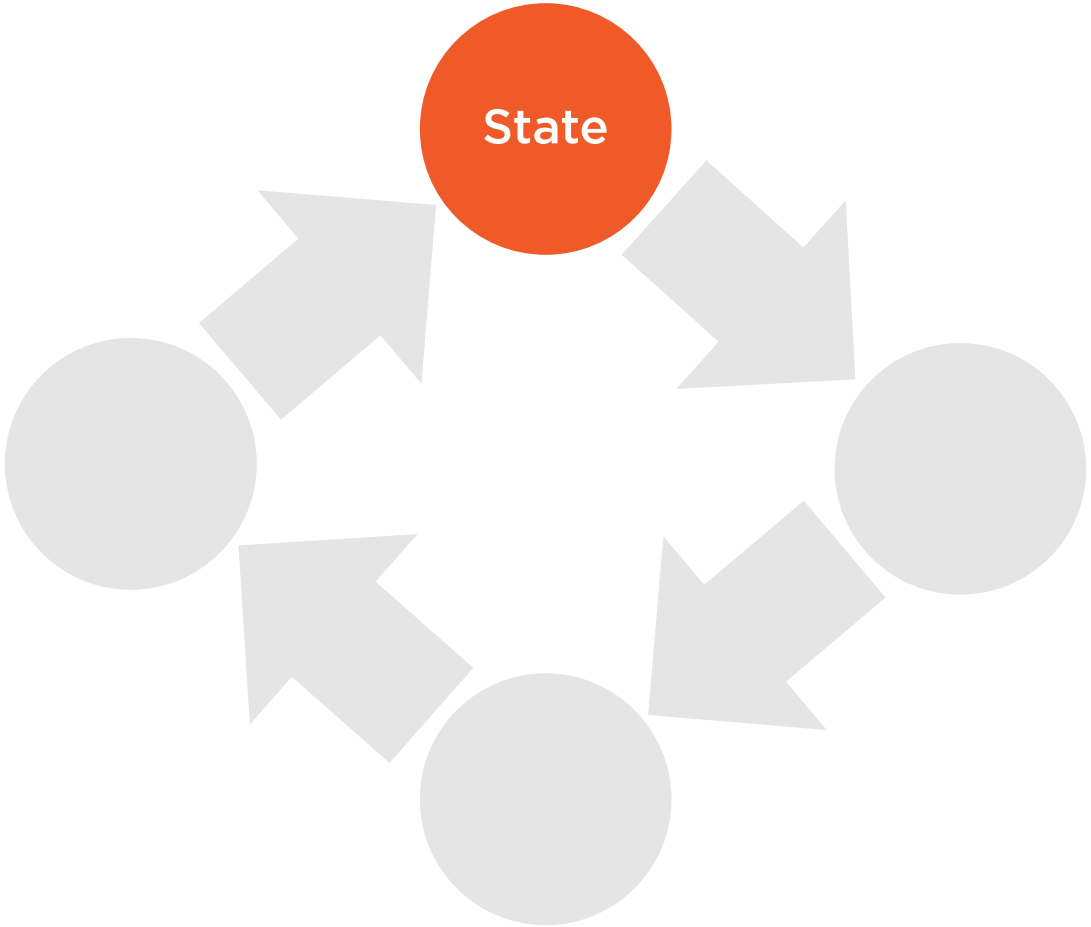
It’s All Just a Function



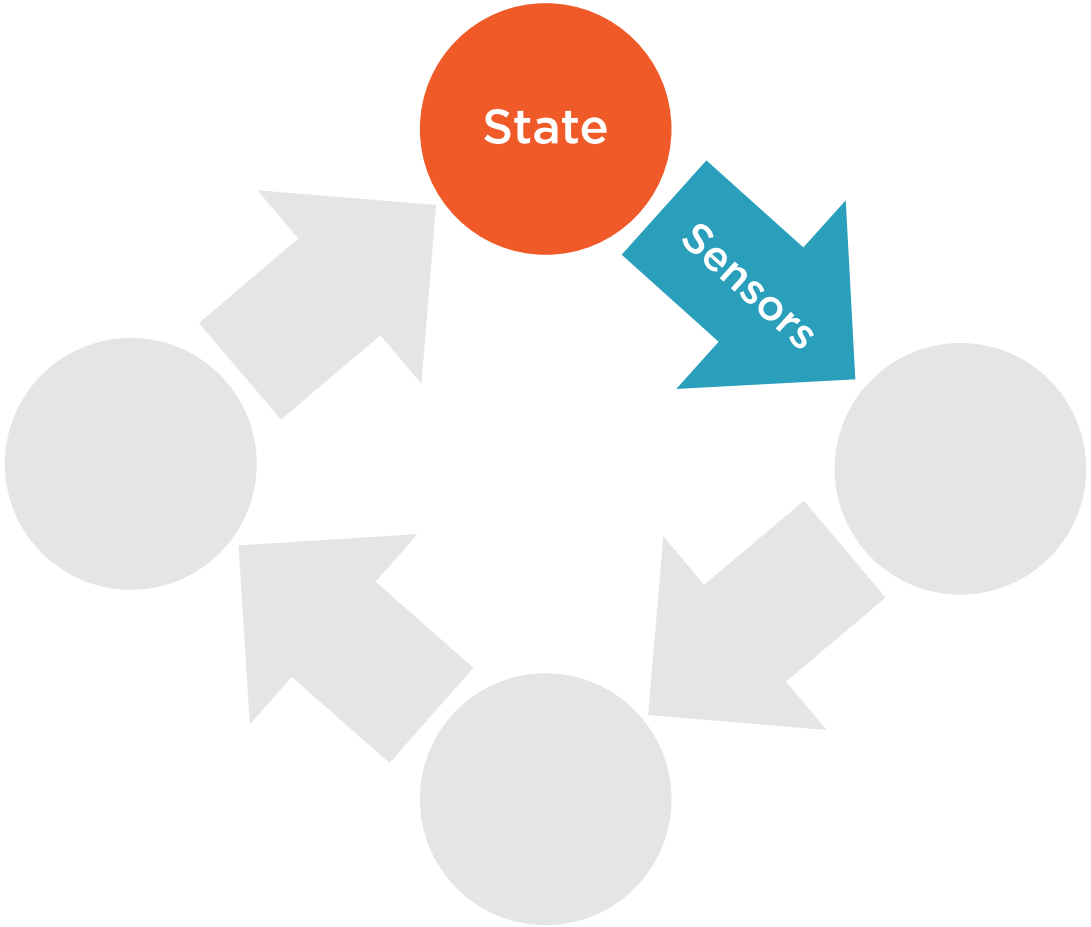
It's All Just a Function



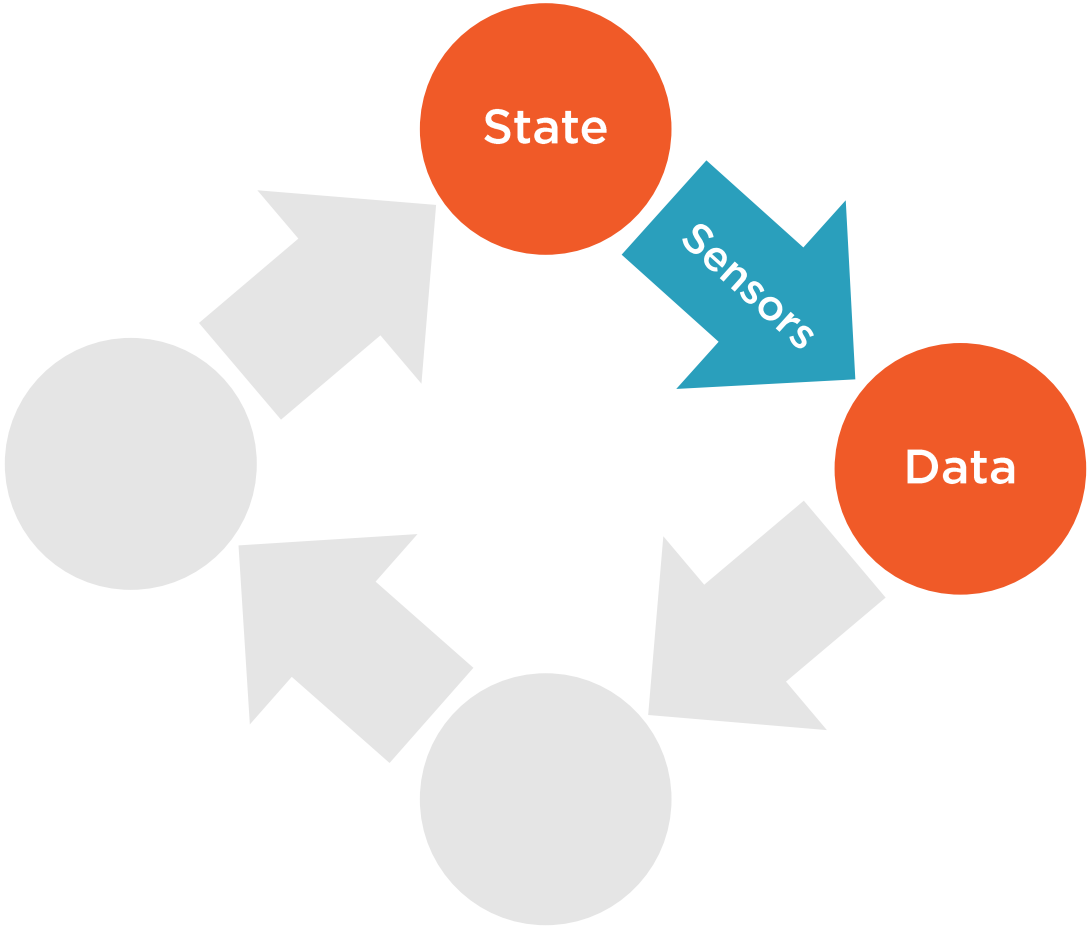
State



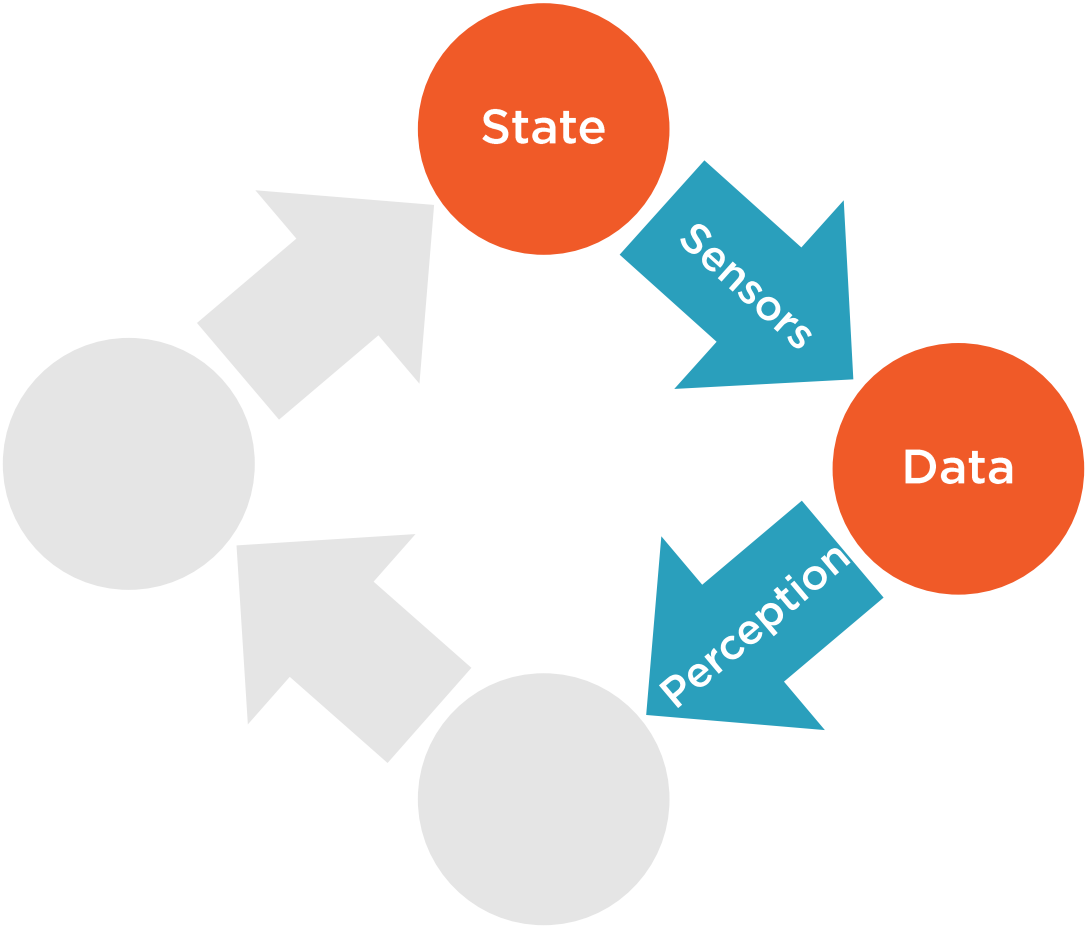
Sensors

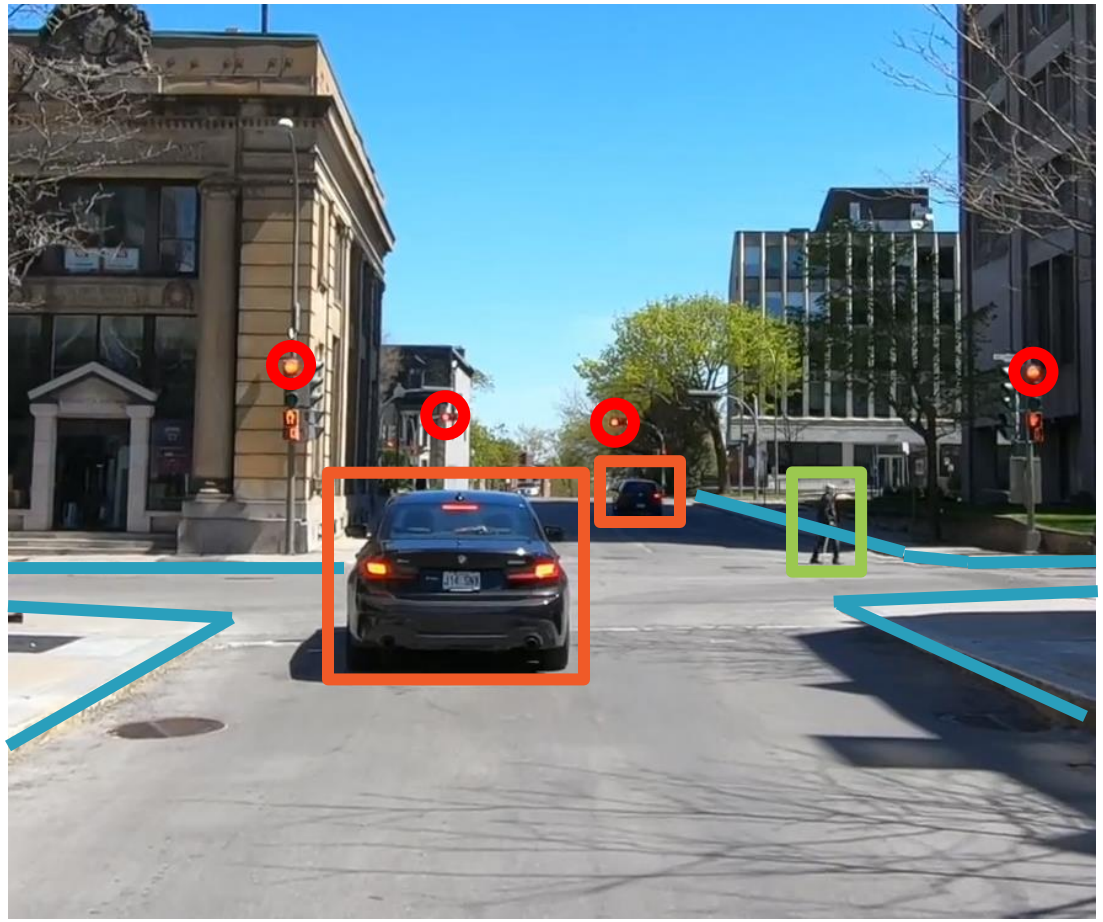


Data

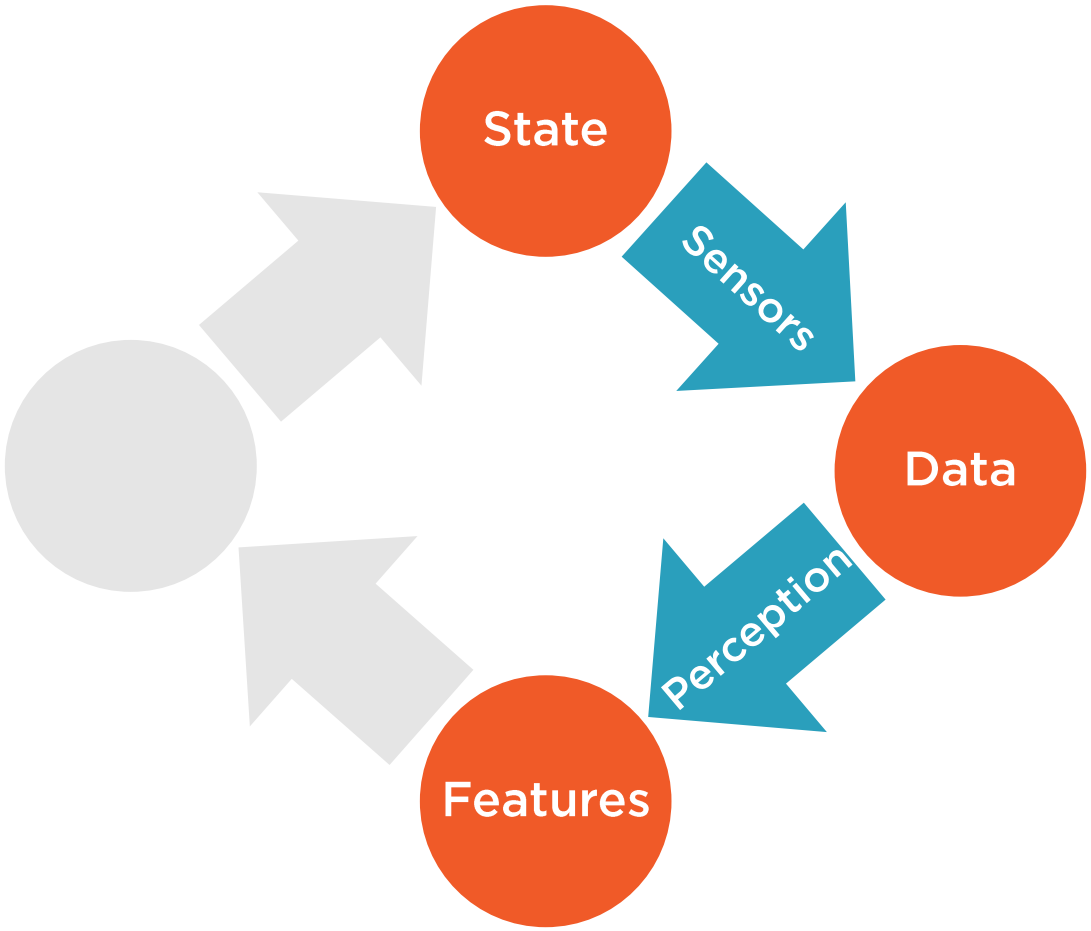


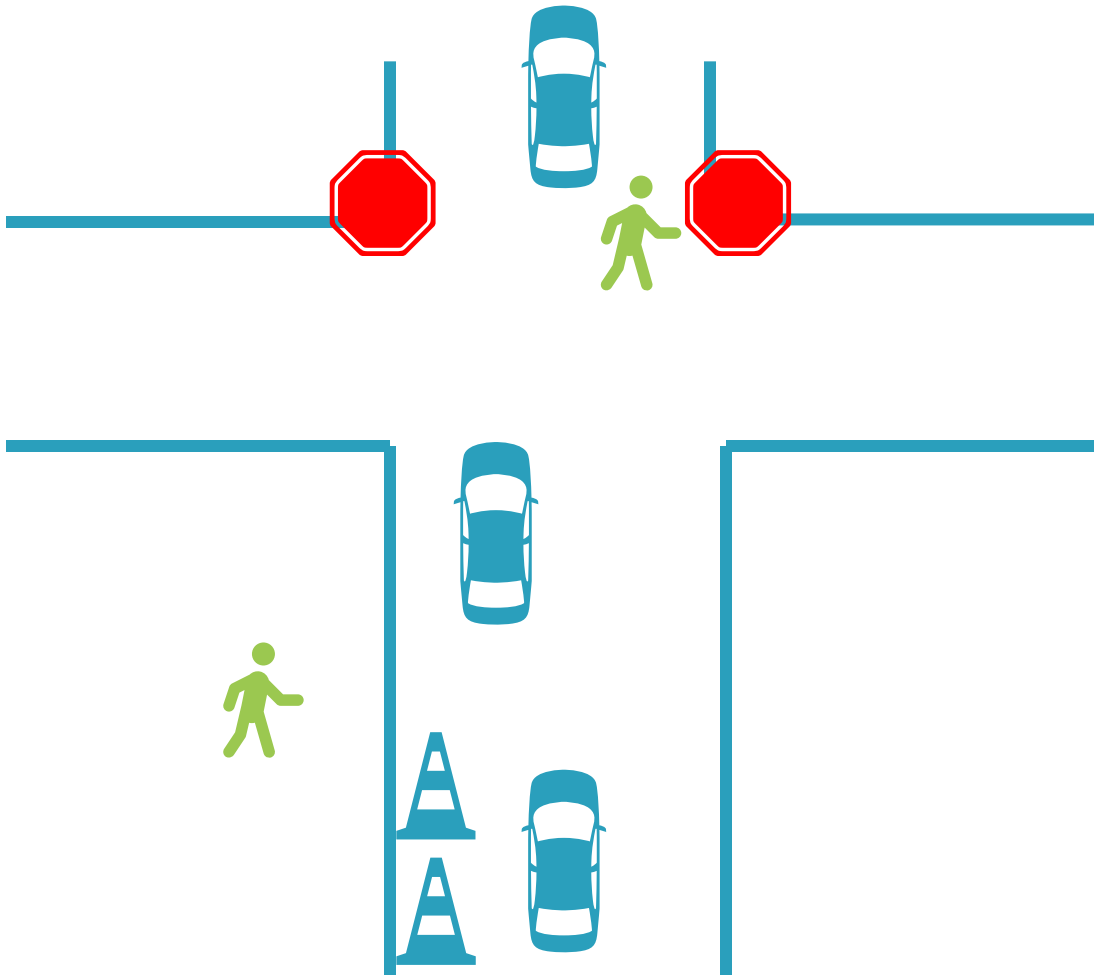
Perception



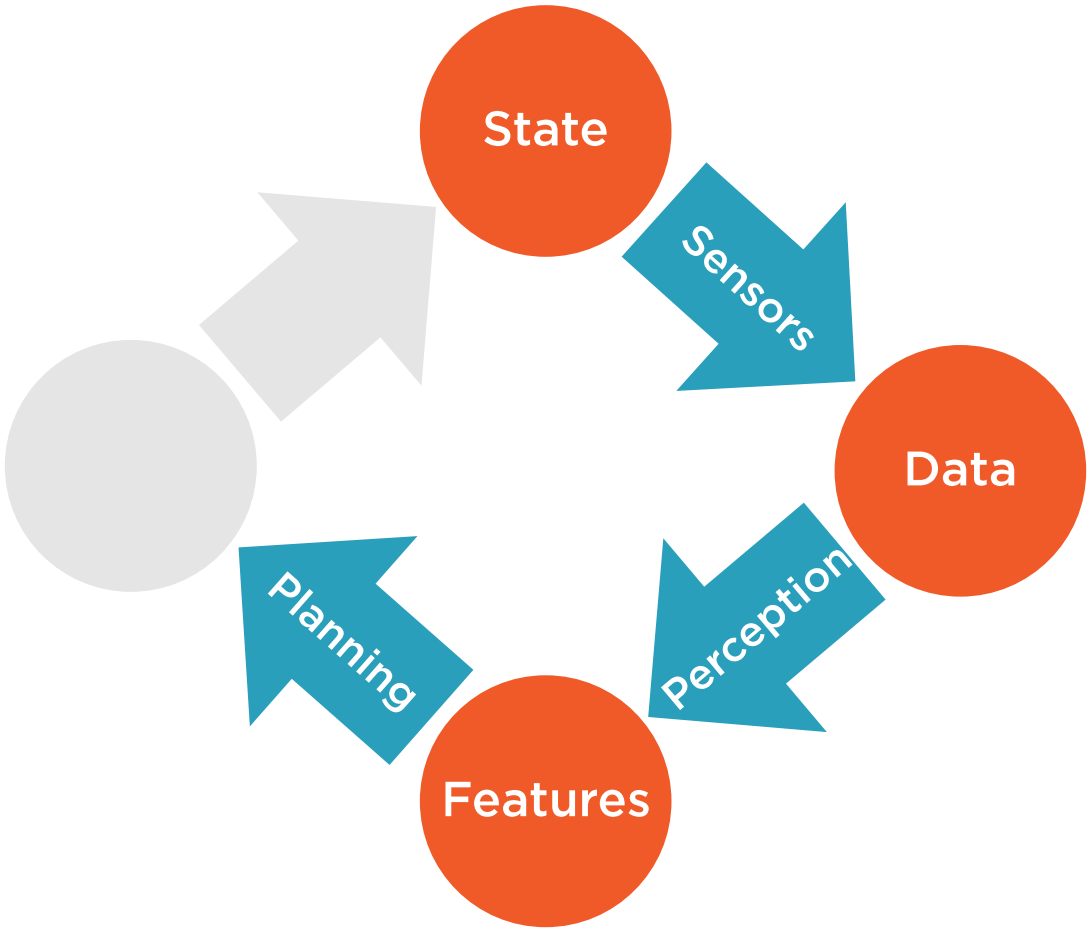


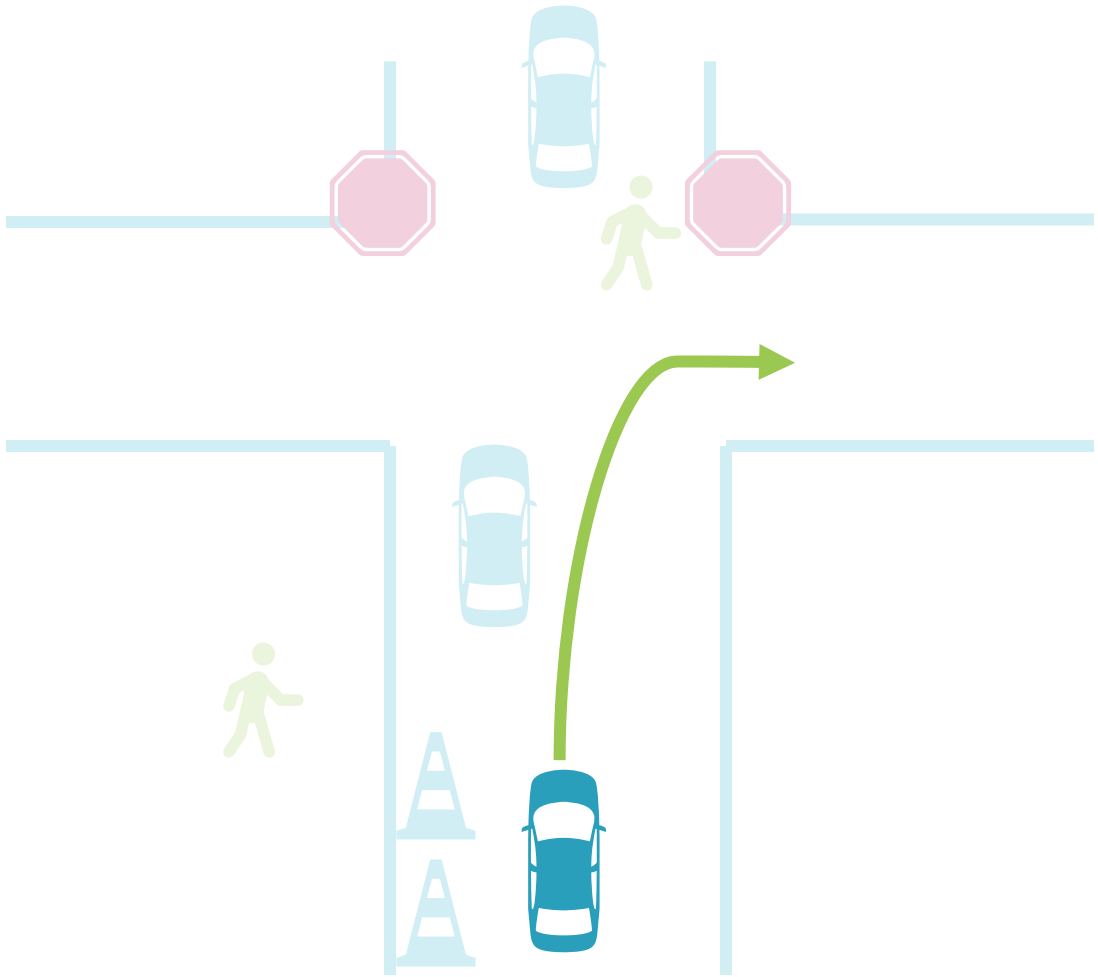
Features





Planning





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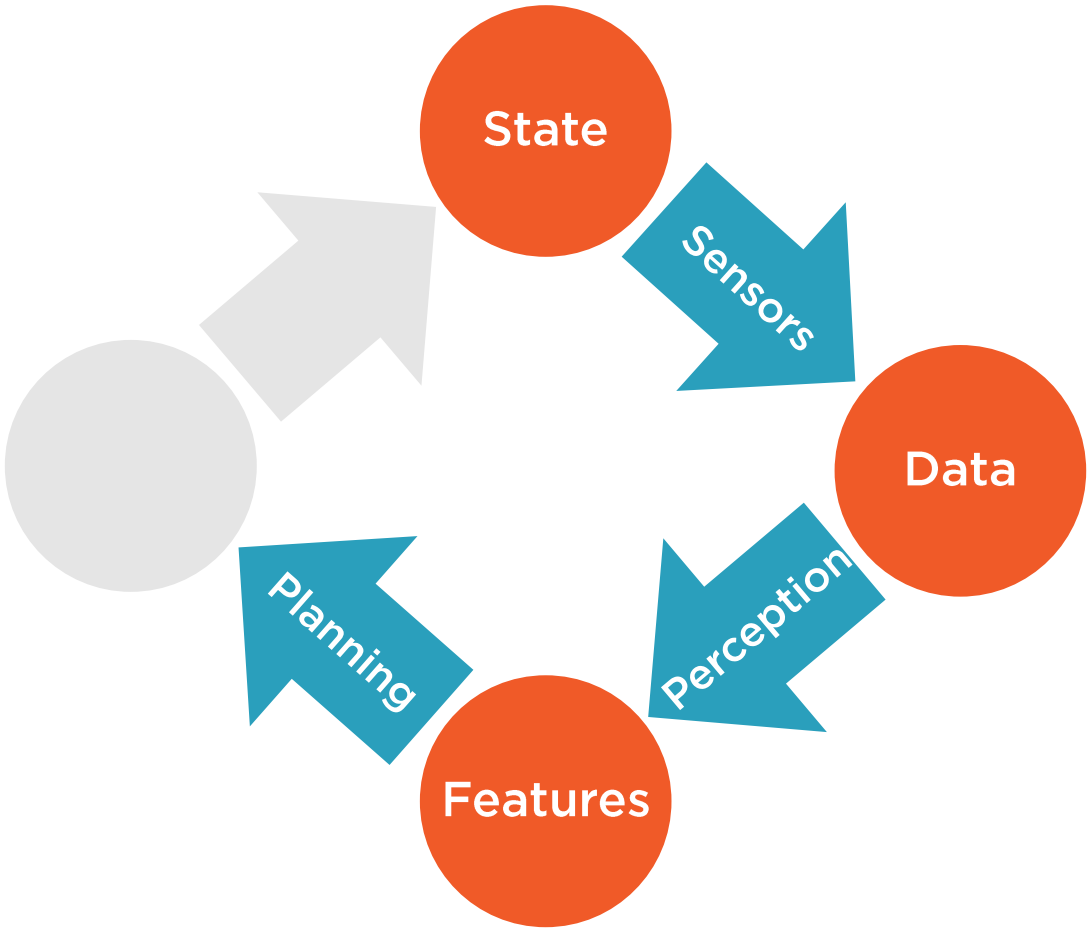
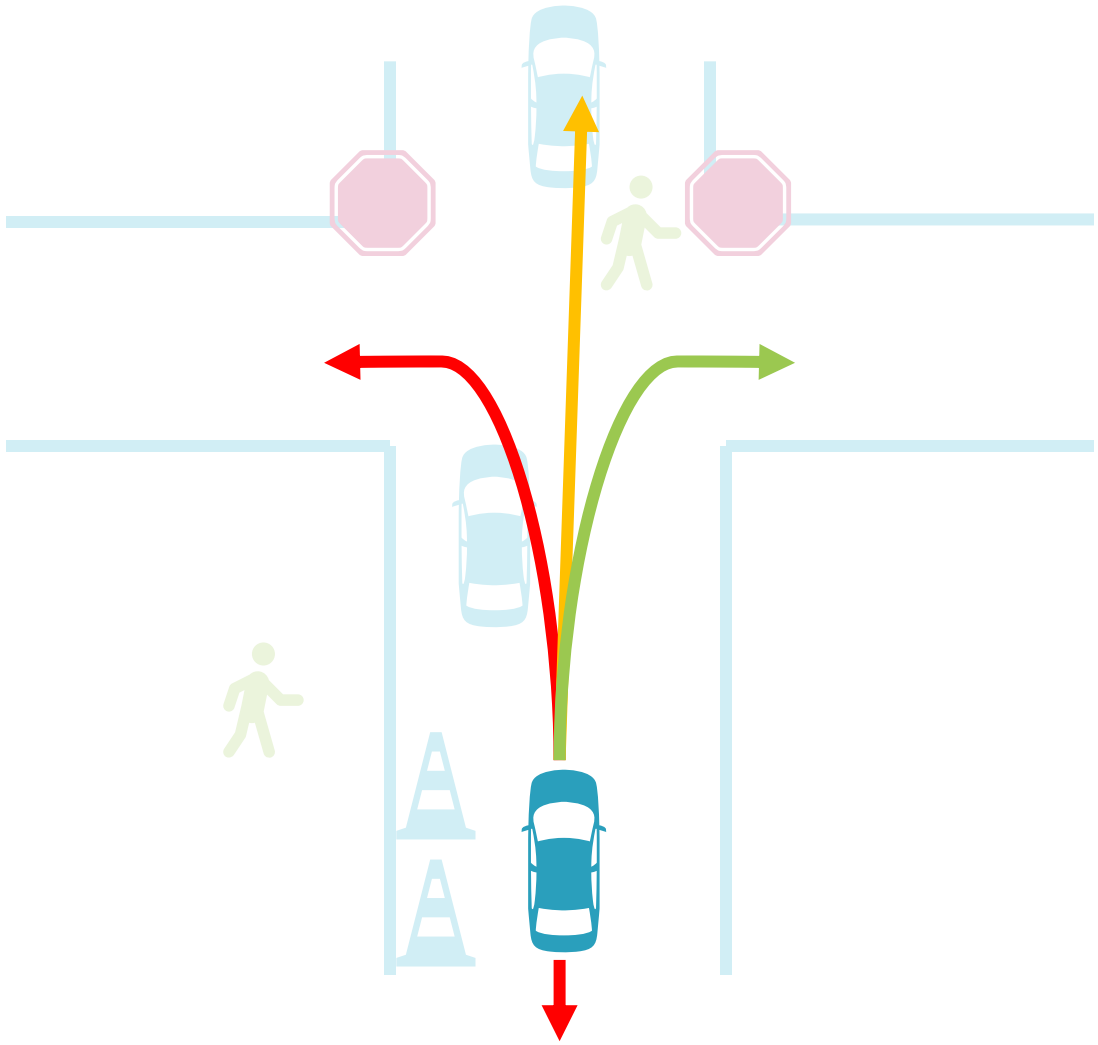
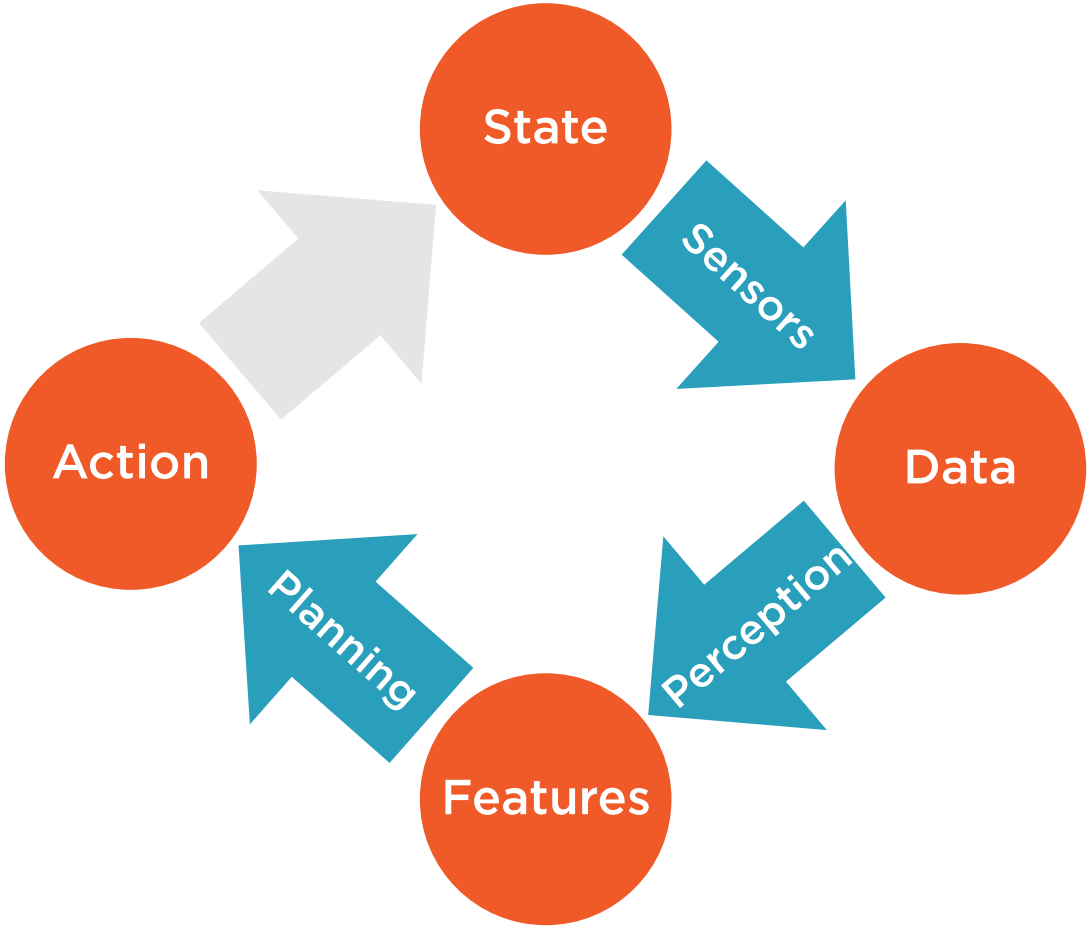
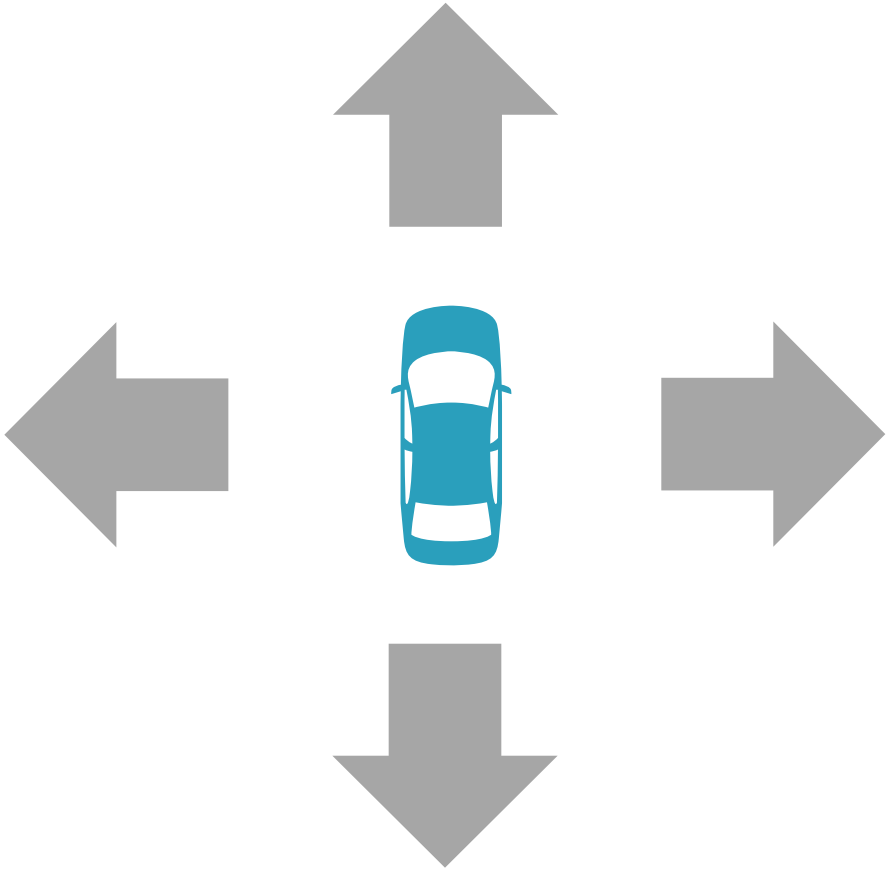




Actions





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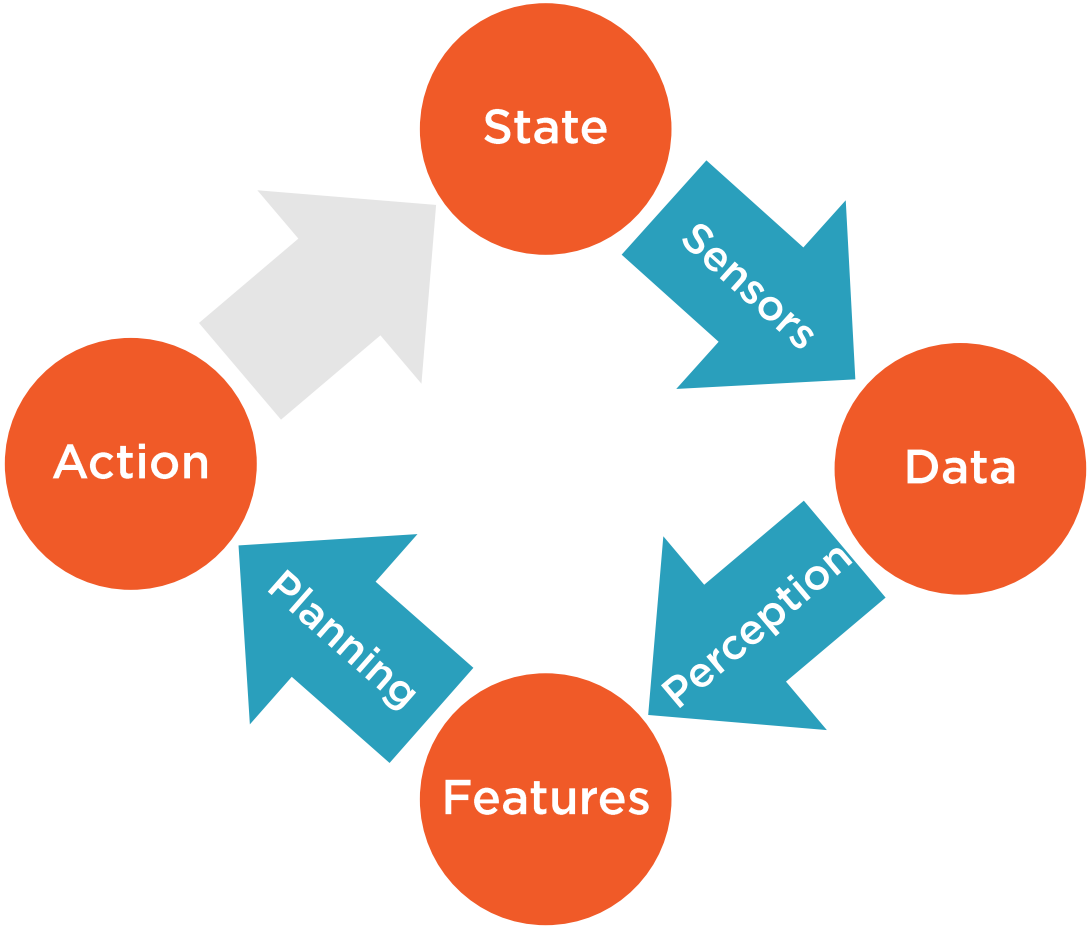
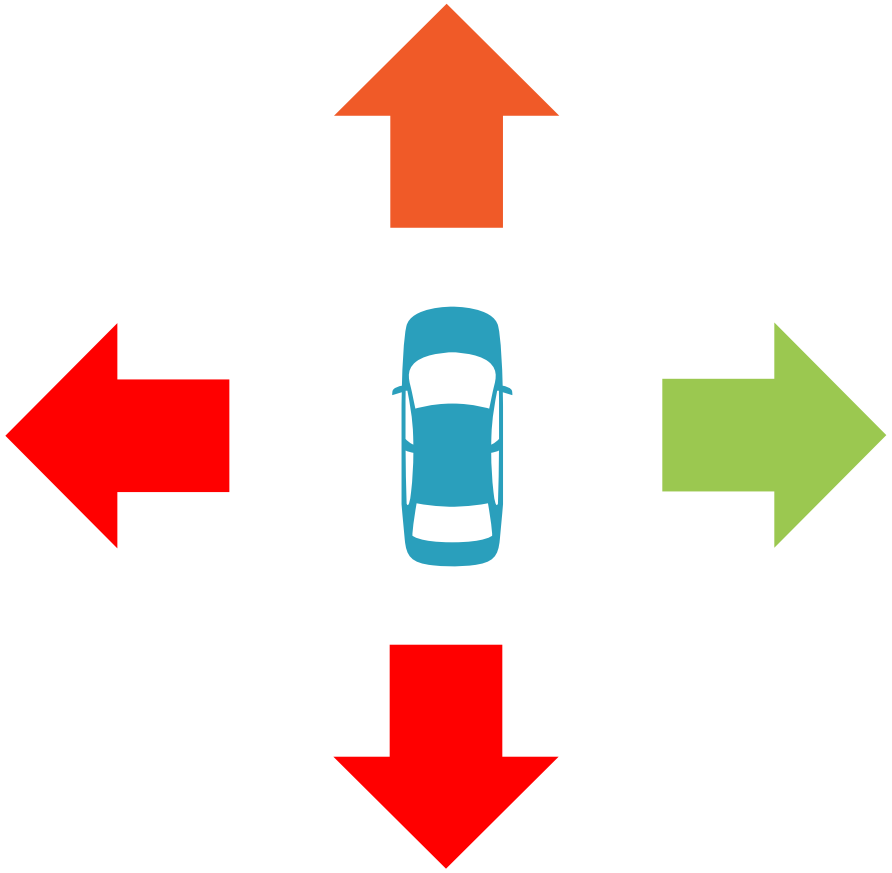
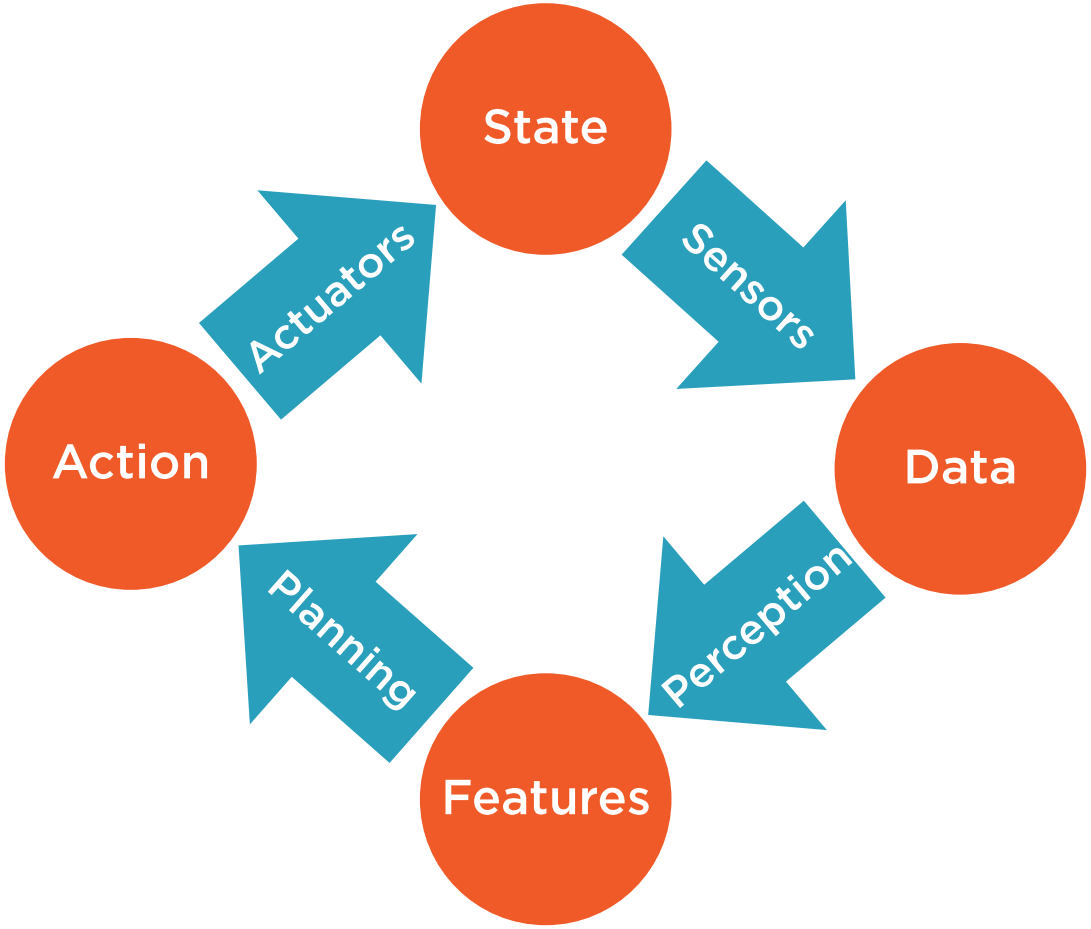
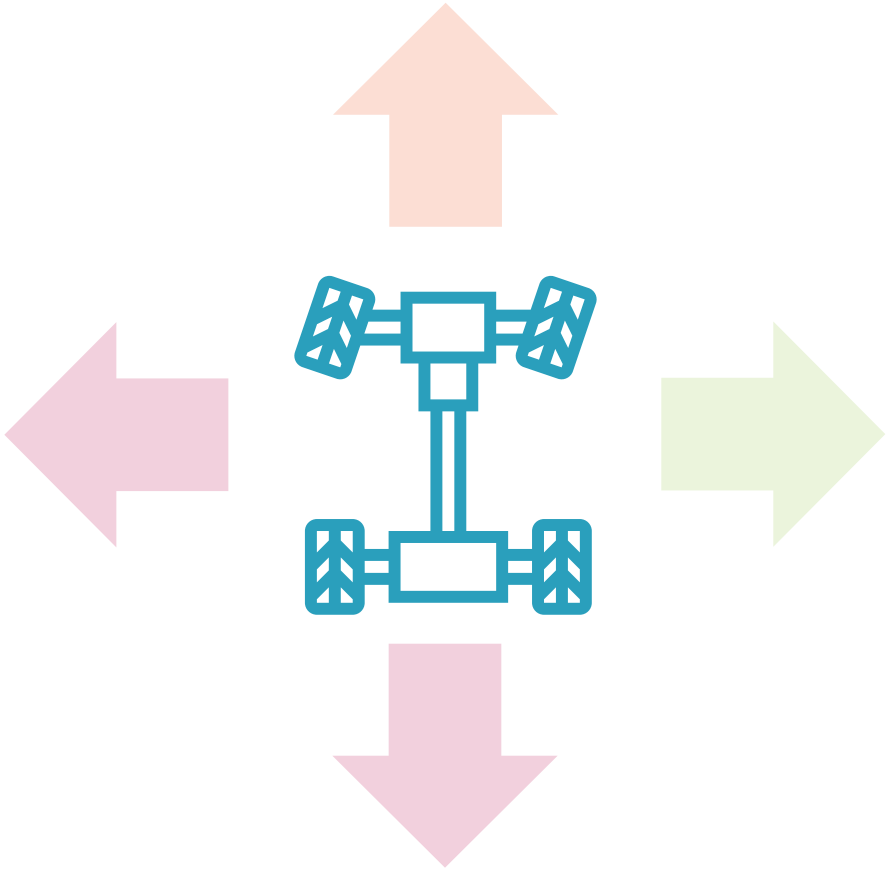


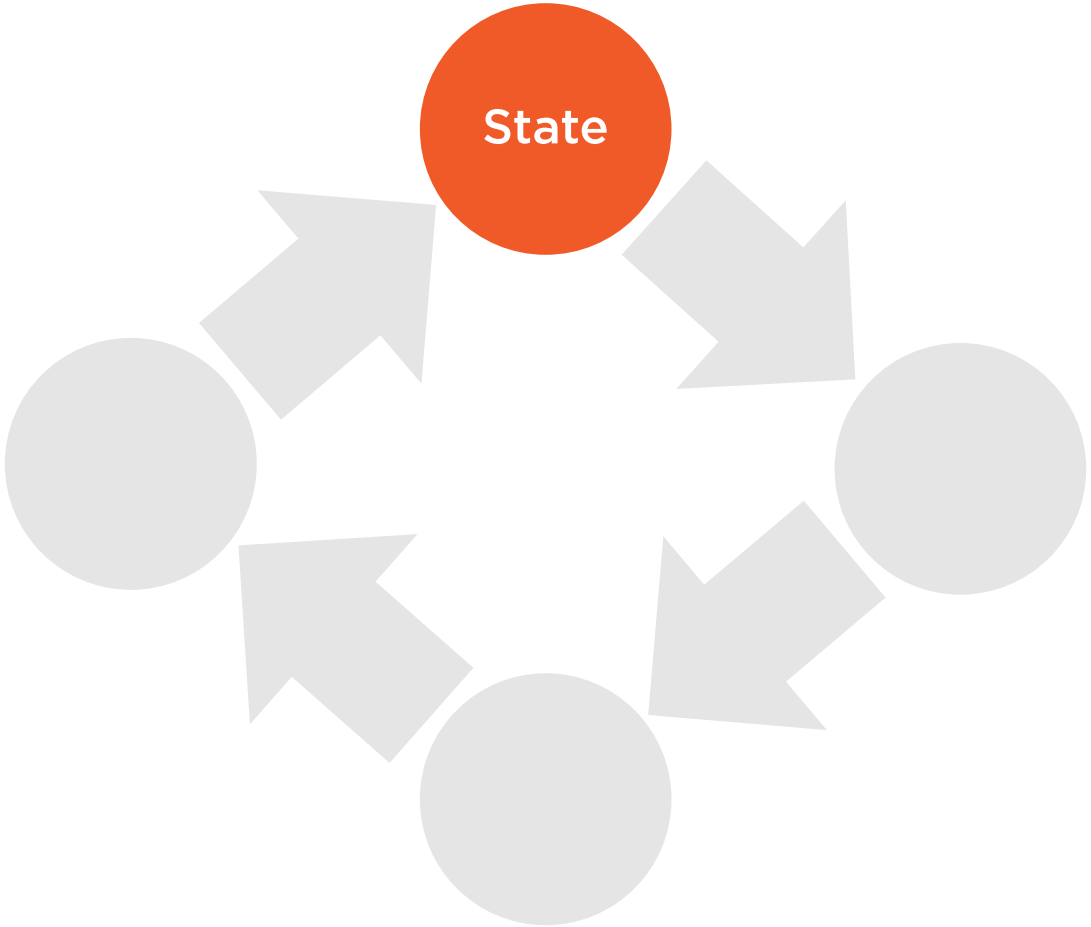


Actuators


State
Sensors
Data
Perception
Features
Planning
Action
Actuators



Feedback



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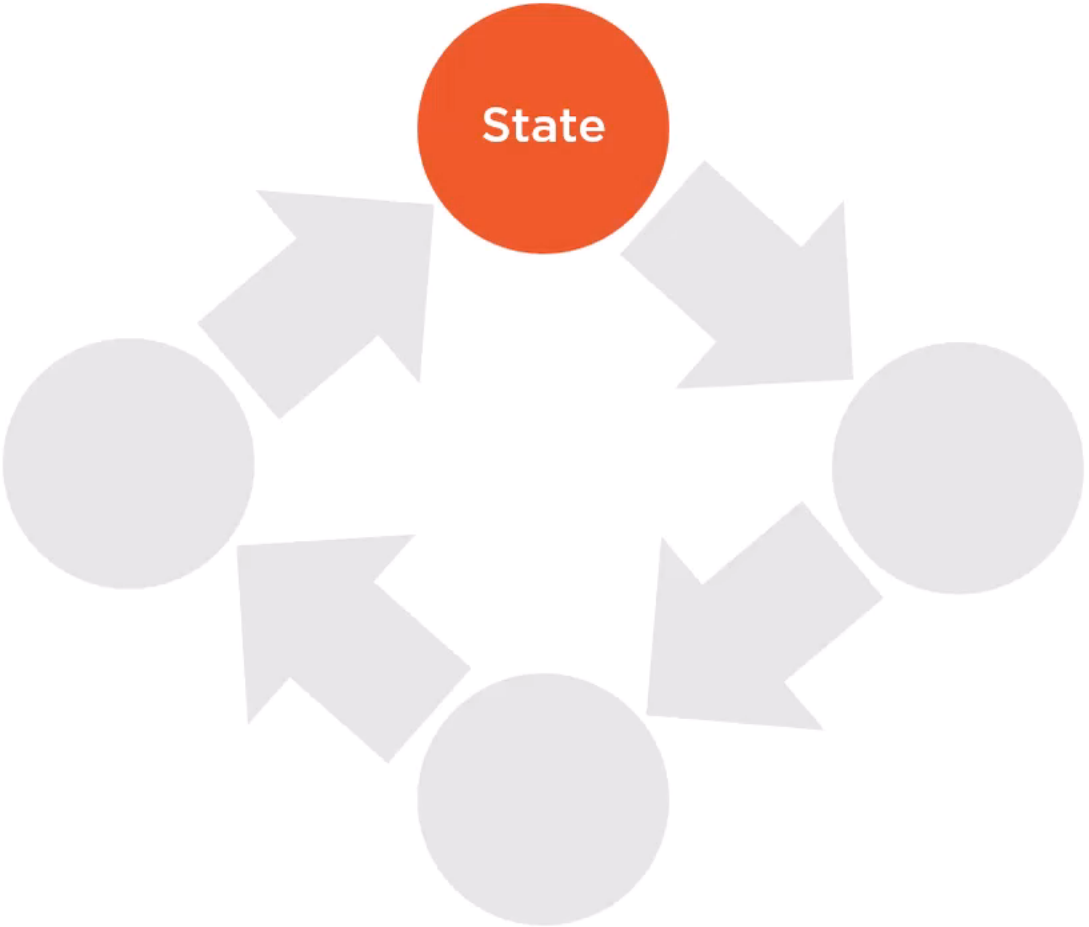
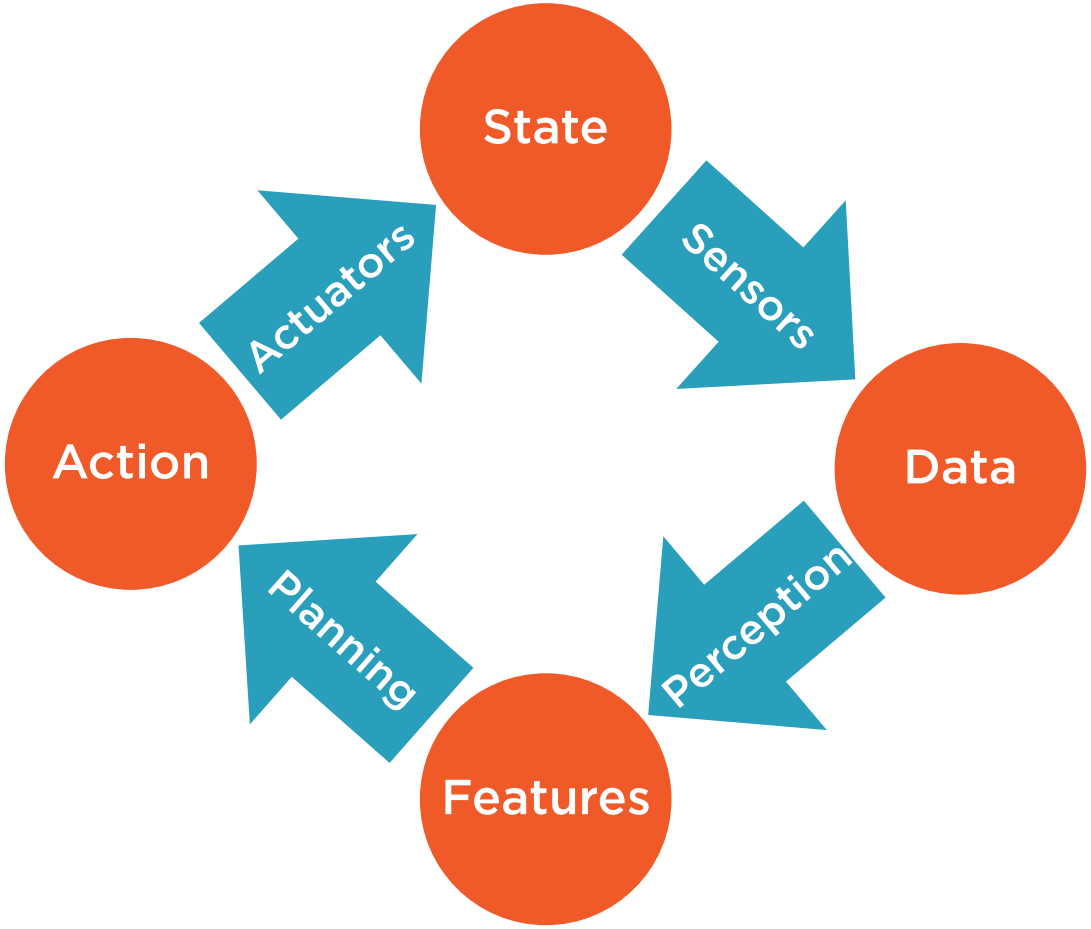


Courtesy of Ford Motor Company

Feedback



Goal: Drive to destination

Sub-goal 1: Drive via Hwy 1

Sub-goal 2: Drive to onramp 1A

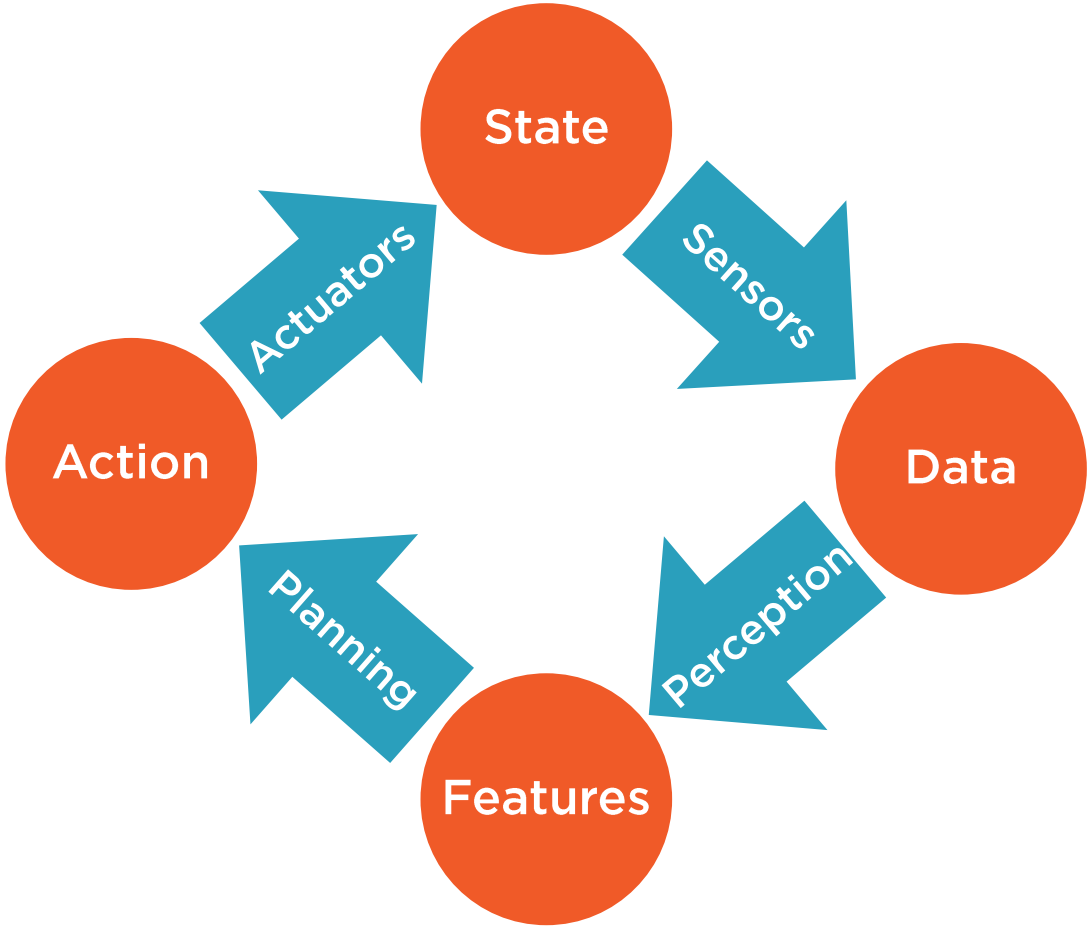
Sub-goal 3: Turn left at onramp

Constraint 1: Stay in lane

Constraint 2: Obey speed limit

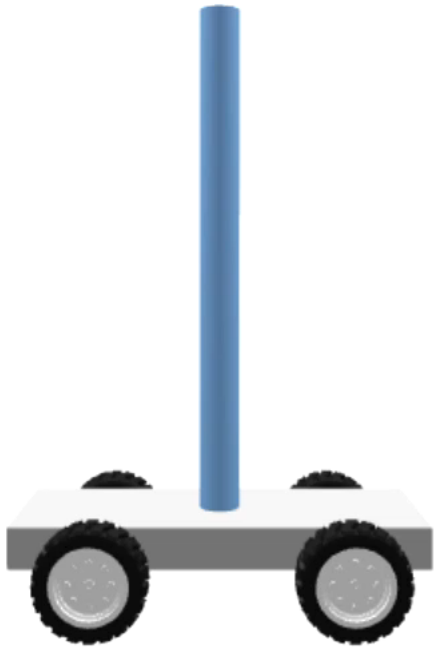
Constraint 3: Avoid obstacles

System Summary



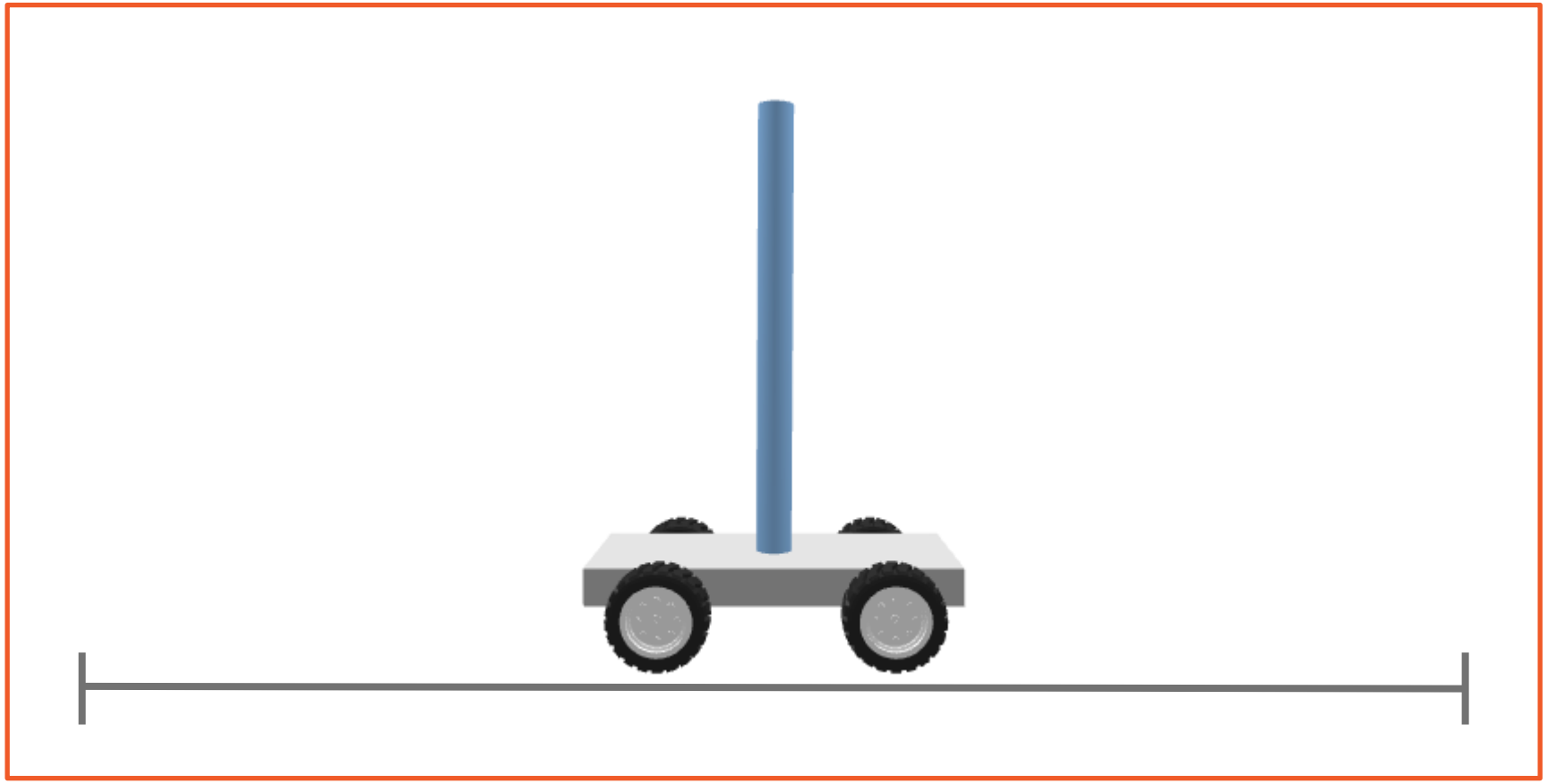
Demo

Demo Overview

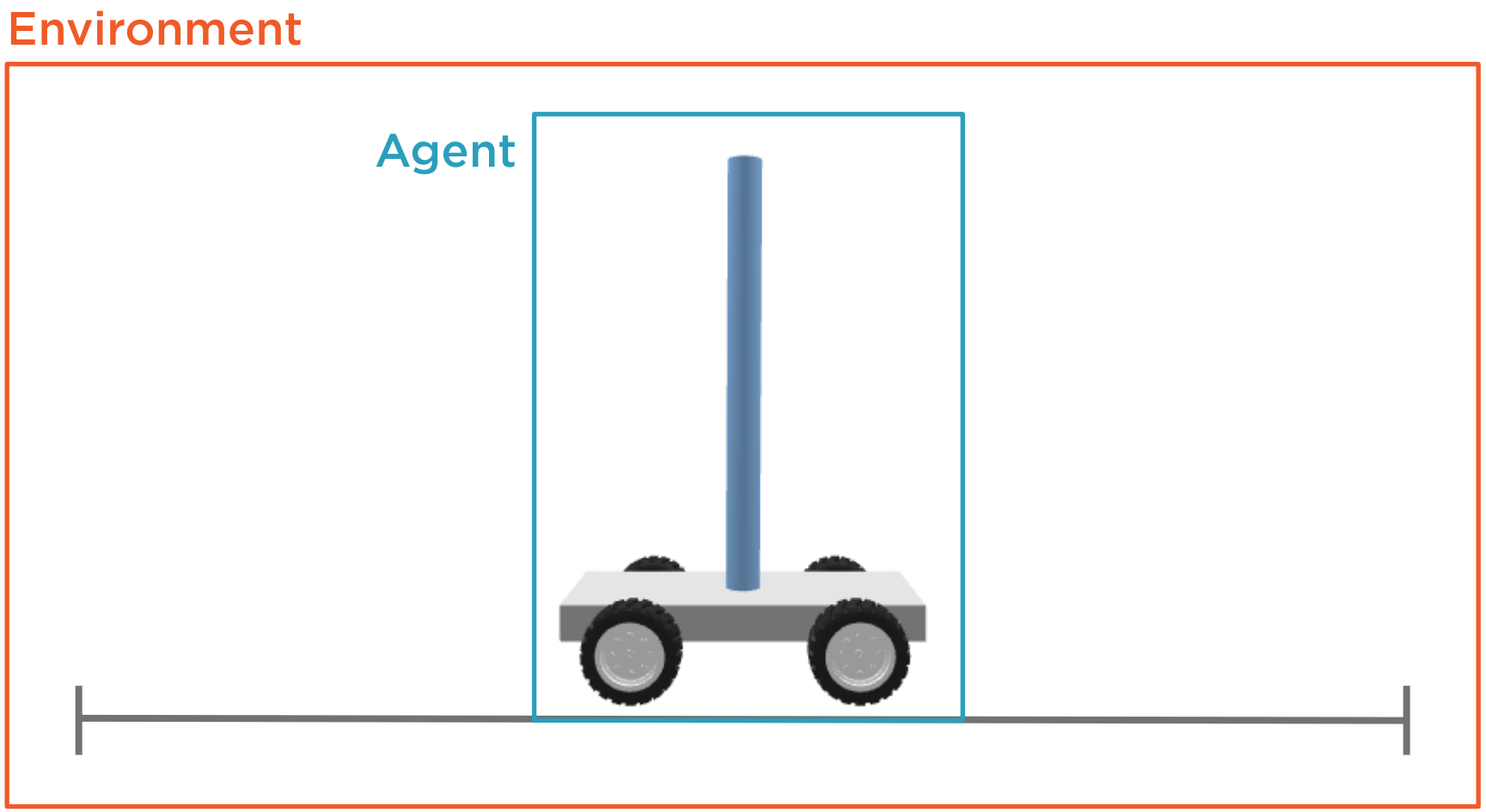


Simulation

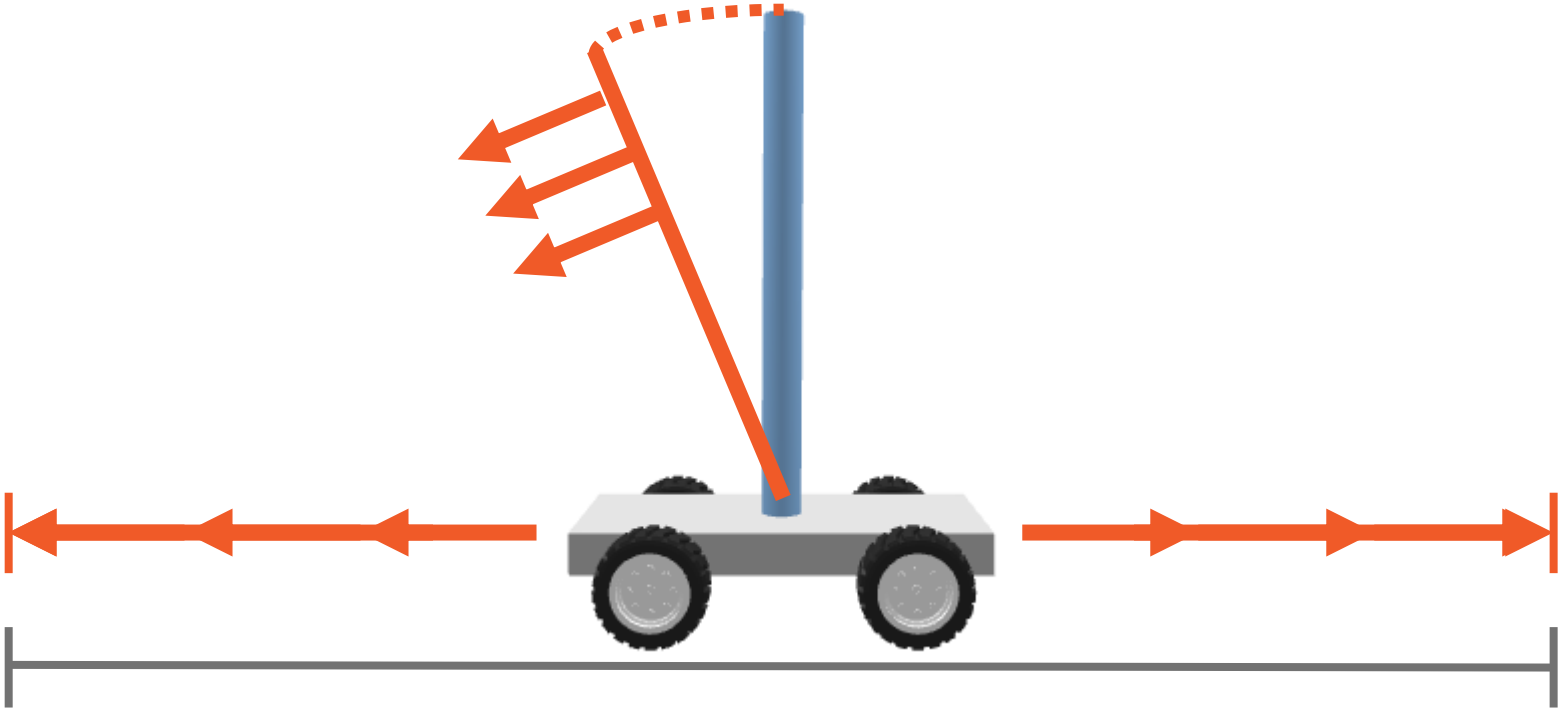
Simulation



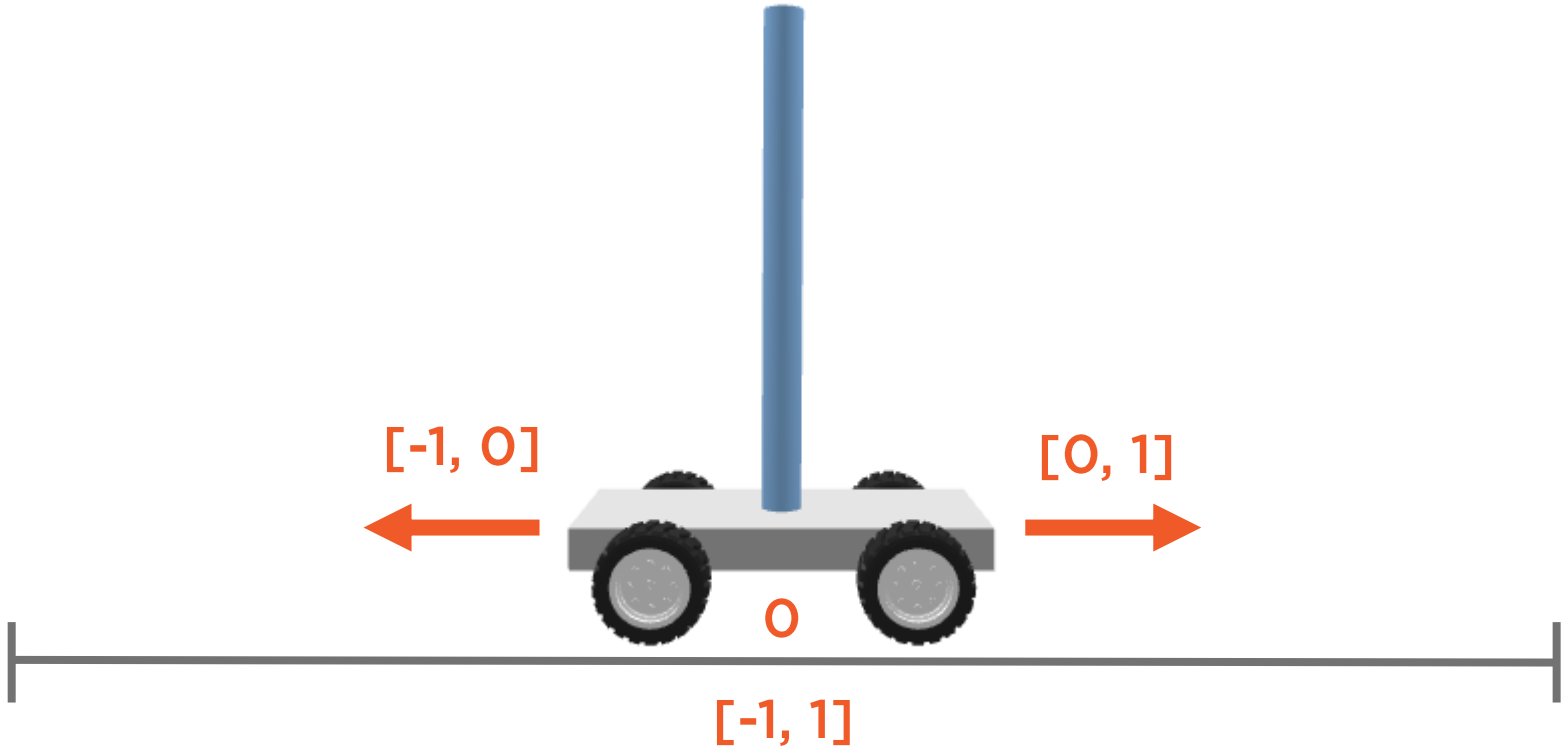
Agent + Environment



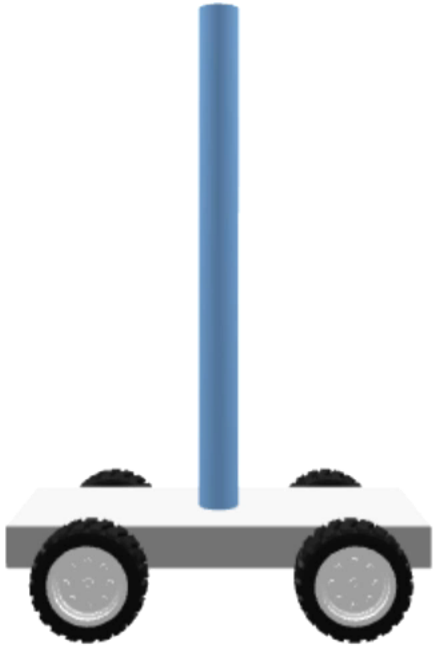
State



Actions



Goal



Demo Options

Microsoft

Bonsai

Bonsai

Bonsai

Search (Ctrl+/)

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems

Settings

- Properties
- Locks

Automation

- Tasks (preview)

Support + troubleshooting

- New support request

Delete

Essentials

JSON View

Resource group (change)
Bonsai

Location
West US 2

Subscription (change)
RenzeConsulting

Subscription ID
[redacted]

Registry
bonsai

Storage
bonsai

Managed resource group
bonsai

Log Analytics Workspace
bonsai

WorkspaceId
[redacted]

Tags (change)
Click here to add tags

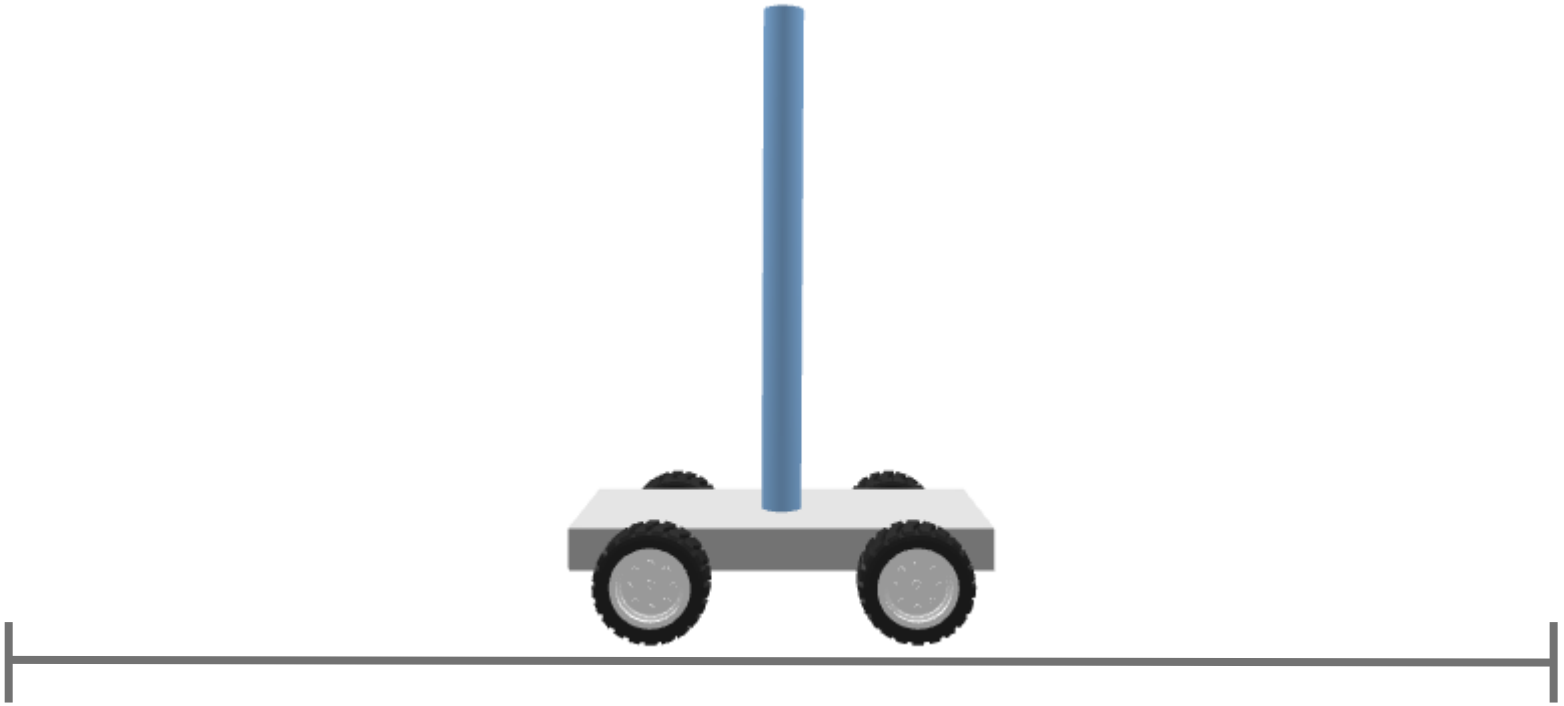
Welcome to Bonsai

Create and optimize intelligence for industrial systems with Deep Reinforcement Learning and Machine Teaching

Launch Bonsai Workspace →

https://github.com/matthewrenze/ai-developers-toolkit/

Constraints



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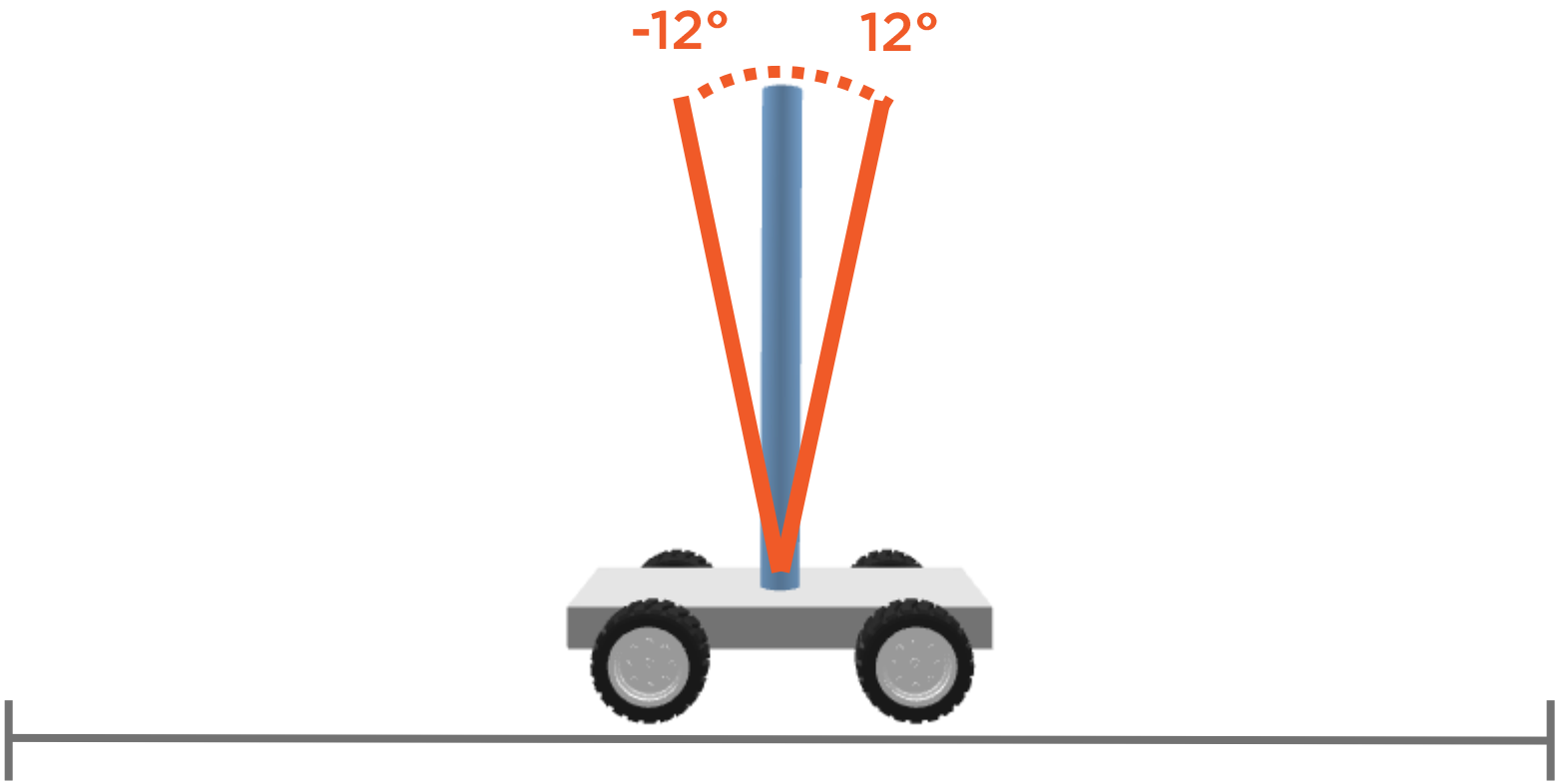

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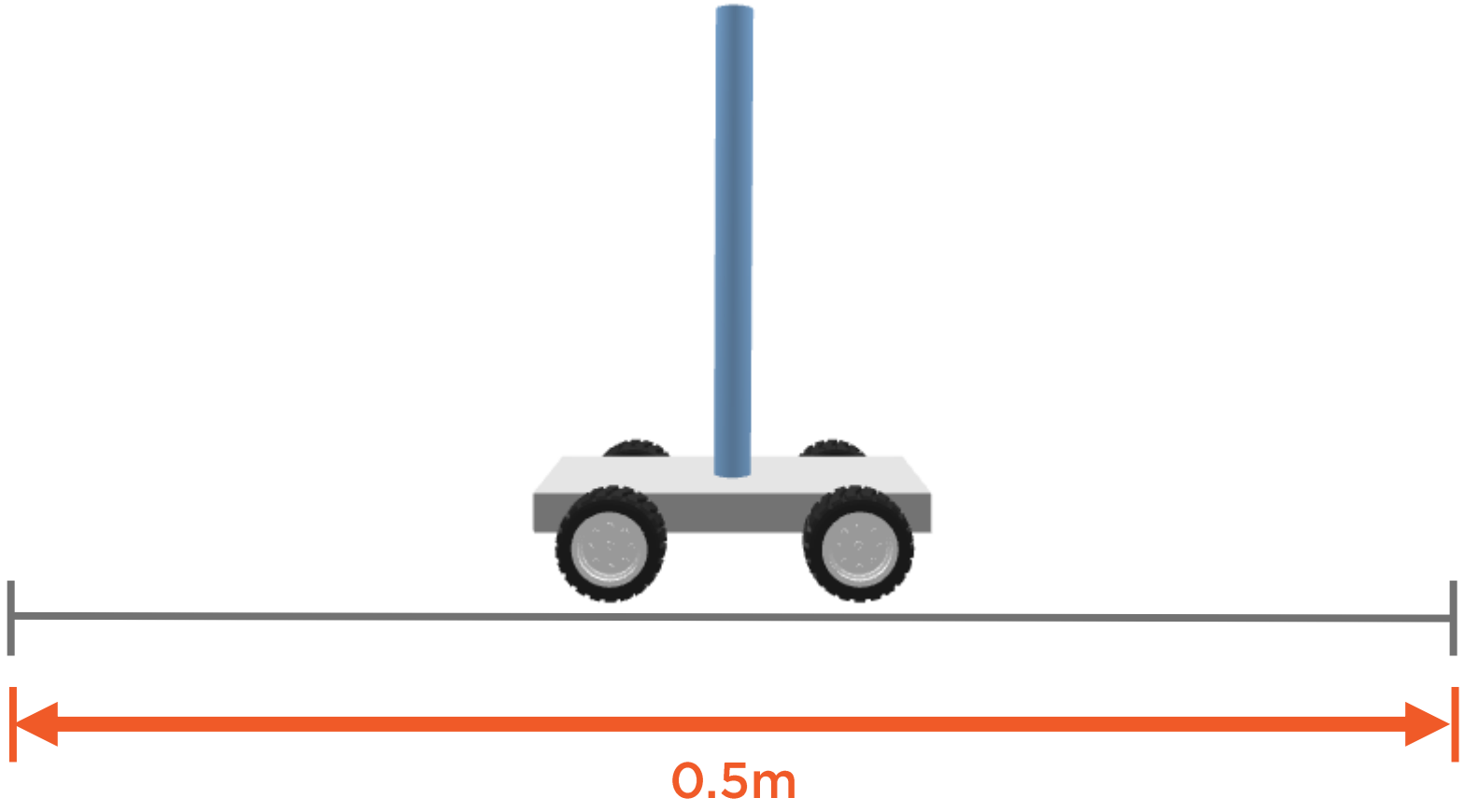

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